

WASP-10b

R-2225

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_110121 · created 2026-08-15T00:22:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455582.57821 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.054
Transit depth (Rp/Rs)^2	2.42 +/- 1.68 [%]
Orbital Inclination (inc)	84.14 +/- 1.97
Airmass coefficient 1 (a1)	0.998 +/- 0.019
Airmass coefficient 2 (a2)	0.002 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#2 - [141, 419]
Optimal Aperture	2.82
Optimal Annulus	7.81
Transit Duration (day)	0.01 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.054 vs 0.1592 (deviation 0.06σ)
Duration fit vs published	0.01 d vs 0.0946 d (deviation 6.51σ)
Mid-time O–C	-0.4 min
Depth SNR	2.9
Residual scatter	1.93 %

Light curve

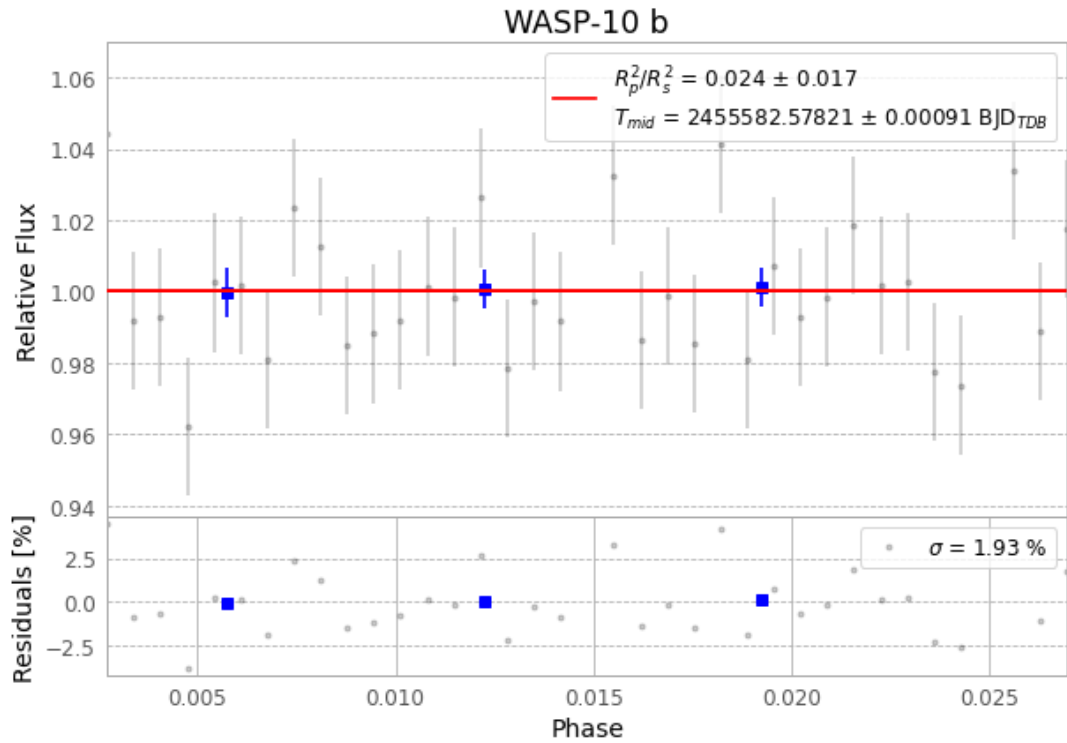


Figure 1 — FinalLightCurve_WASP-10 b_2011-01-20.png (SHA-256 ed3ef434e974043d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 128a232bbeb47f3d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

36 FITS frames (24 MB), WASP-10110121020613.FITS → WASP-10110121035414.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-110121.log (1512525e029a...), FinalLightCurve_WASP-10 b_2011-01-20.png (ed3ef434e974...), FinalLightCurve_WASP-10 b_2011-01-20.csv (8c39e3bba2a2...)

Compute resources

Wall time 87.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 00:22Z.